The documentation and process conversion measures necessary to comply with this revision shall be completed by 7 January 1990

INCH-POUND

MIL-S-19500/308B
7 April 1989
SUPERSEDING
MIL-S-19500/308A
9 June 1970

MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, DIODE, SILICON, POWER RECTIFIER, FAST-RECOVERY
TYPES 1N3909, 1N3910, 1N3911, 1N3912, 1N3913, R AND A VERSIONS
JAN, JANTX, AND JANTXV

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for silicon, semiconductor fast recovery, power rectifier diodes. Three levels of product assurance are provided for each device type as specified in MIL-S-19500.

1.2 Physical dimensions. See figure 1 (DO-5).

1.3 Maximum ratings.

Types	V_R	V_{RWM}	I_O 1/ T_C = 100°C	i_f (surge) at 1/120 s T_C = 100°C	trr
	V dc	v(pk)	A dc	a dc	ns
1N3909, R	50	50	30	300	200
1N3909A, AR	50	50	50	400	150
1N3910, R	100	100	30	300	200
1N3910A, AR	100	100	50	400	150
1N3911, R	200	200	30	300	200
1N3911A, AR	200	200	50	400	150
1N3912, R	300	300	30	300	200
1N3912A, AR	300	300	50	400	150
1N3913, R	400	400	30	300	200
1N3913A, AR	400	400	50	400	150

1/ Derate linearly, 2 percent of I_O per °C for T_C > 100°C.

Storage temperature: T_C = -65°C to +175°C,
Operating temperature: T_C = -65°C to +150°C,
Barometric pressure reduced (altitude operation): 8 mmHg.

1.4 Primary electrical characteristics. $R_{\theta JC}$ = .8°C/W maximum.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Laboratory Command, ATTN: SLCET-R-S, Fort Monmouth, NJ 07703-5302 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A
FSC 5961

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARDS

FEDERAL

FED-STD-H28 - Screw-Thread Standards for Federal Services.

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets, or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be in accordance with MIL-S-19500, and as specified herein. Lot accumulation period shall be three months in lieu of six weeks.

3.2 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-S-19500.

3.3 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-S-19500, and figure 1 herein.

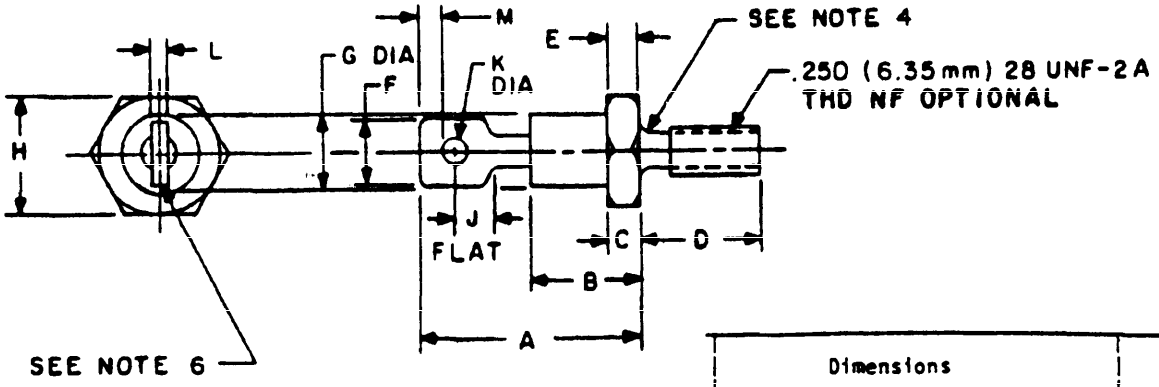
3.4 Marking. Marking shall be in accordance with MIL-S-19500, except at the option of the manufacturer, the following marking may be omitted from the body of the device:

a. Country of origin.

b. Manufacturer's identification.

3.5 Polarity. The polarity shall be indicated by a graphic symbol with the arrow pointing toward the negative end for forward bias. The reversed units shall also be marked with an R following the last digit in the type number.

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Dimensions					Notes
Dim	Inches		Millimeters		
	Min	Max	Min	Max	
A	.750	1.000	19.05	25.40	
B	---	.450	---	11.43	
C	.115	.200	2.92	5.08	
D	.422	.453	10.72	11.51	
E	.060	---	1.52	---	6

Dimensions					Notes
Dim	Inches		Millimeters		
	Min	Max	Min	Max	
F	.250	.375	6.35	.953	6
G	---	.667	---	16.94	
H	.669	.688	16.99	17.48	
J	.156	---	3.96	---	
K	.140	.175	3.56	4.45	
L	.030	.080	.76	2.03	
M	.030	- - -	.77	- - -	

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Units must not be damaged by torque of 30 inch-pounds applied to .250-28 UNF-2B nut assembled on thread.
4. Diameter of unthreaded portion .249 (6.32 mm) max and .220 (5.59 mm) minimum.
5. Complete threads to extend to within 2-1/2 threads of seating plane.
6. Angular orientation for this terminal is underlined, however the major surfaces over dimension F shall be flat and the minimum distance from the hole to any point on the periphery shall be .030 inches (0.76 mm) outside dimension J.
7. Max pitch diameter of plated threads shall be basic pitch diameter .2268 (5.76 mm) reference FED-STD-H28 (screw thread standards for federal services).
8. A chamfer or undercut on one or both ends of the hex portion is optional; minimum base diameter at seating plane .600 inch (15.24 mm).
9. Reversed (anode to stud) units shall be marked with an R following the last digit in the type number.

FIGURE 1. Physical dimensons.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-S-19500, and as specified herein. Tests in either polarity shall be sufficient to obtain qualification approval of both polarities.

4.3 Screening (JANTX and JANTXV levels only). Screening shall be in accordance with MIL-S-19500 (table II), and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table II of MIL-S-19500)	Measurement
	JANTX and JANTXV levels
1/	Surge, see 4.3.1 Thermal response, see 4.3.2
4	Not applicable
9	V_{F2} and I_{R1} 2/
10	MIL-STD-750, method 1038 test condition A, t = 48 hours
11 3/	Subgroup 2 of table I herein V_{F2} and I_{R1}; ΔV_{F2} = ±0.1 v(pk); ΔI_{R1} = ±5 µA dc or 100% from the initial value, whichever is greater.
12	Not applicable
13	Not applicable

1/ Surge shall precede thermal response. These tests shall be performed anytime after screen 3 and before screen 9.
2/ I_{R1} measurement shall not be indicative of an open condition.
3/ PDA of screen 13 shall apply to screen 11.

4.3.1 Surge current. Surge current, see MIL-STD-750, method 4066. I_O = 0; $V_{RM}(w)$ = 0; Non-A version, I_{FSM} = 400 A; A version, I_{FSM} = 500 A; six surges; T_A = 25°C; t_p = 8.3 ms

4.3.2 Thermal response (ΔV_F measurements). The delta V_F measurements shall be performed in accordance with MIL-STD-750, method 3101. The delta V_F conditions and maximum V_F limit shall be derived by each vendor. The chosen ΔV_F measurement and conditions for each device in the qualification lot shall be submitted in the qualification report and a thermal response curve shall be plotted. The chosen ΔV_F value shall be considered final after the manufacturer has had the opportunity to test five consecutive lots. Parameter t_p = 250 ms. Heating current (I_H) ≥ rated I_O.

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4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-S-19500, and table I herein. (End-point electrical measurements shall be in accordance with the applicable steps of table V herein.)

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table IVb (JANTX and JANTXV) of MIL-S-19500, and table II herein. Electrical measurements (end points) and delta requirements shall be in accordance with the applicable steps of table V herein.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table V of MIL-S-19500, and table III herein. Electrical measurements (end points) and delta requirements shall be in accordance with the applicable steps of table V herein.

4.4.4 Group E inspection. Group E inspection shall be in accordance with table IV herein.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows:

4.5.1 DC intermittent operation life. A cycle shall consist of an "on" period, when power is applied suddenly, not gradually, to the device for the time necessary to achieve a delta case temperature of 85°C +15°C, -5°C followed by an "off" period, when the power is suddenly removed, for cooling. Auxiliary (forced) cooling is permitted during the "off" period only. The heating time is such that 30 s $\leq t_{heating} \leq$ 60 s. $P = V_F \times I_F$ or $P = V_{F(pk)} \times I_{av}$ if using sine wave current. DC full wave current (or equivalent half sine wave current) shall be used for the power required during the "on" period and equivalent half sine wave is permissable.

Within the time interval of 50 cycles before to 500 cycles after the termination of the test, the sample units shall be removed from the specified test conditions and allowed to reach room ambient conditions. Specified end-point measurements for qualification and quality conformance inspections shall be completed within 96 hours after removal of sample units from the specified test conditions. Additional readings may be taken at the discretion of the manufacturer.

4.5.2 Reverse recovery time. The reverse recovery time shall be measured in the circuit of figure 2 or equivalent. Care should be exercised to minimize stray inductances in the test circuit and to ensure the total resistance of the reverse current loop can be adjusted sufficiently low so that more than 2 amperes will flow if not blocked by the diode being tested. Switch SW shall be activated and the regulated voltage source adjusted to achieve the following characteristics of the waveform.

(1) The di/dt shall be the specified value between the forward 0.5 ampere point and the reverse 0.2 ampere point.

(2) The i_r(rec) shall be the maximum value obtainable, except when it exceeds 2 amperes.

The reverse-recovery time shall then be determined from the current waveform as shown on figure 3.

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TABLE I. Group A inspection.

Inspection	MIL-STD-750		LTPD 1/	Symbol	Limits		Unit
	Method	Conditions			Min	Max	
Subgroup 1							
Visual and mechanical inspection	2071						
Subgroup 2							
Forward voltage	4011	$I_F = I_{FSM}$ $t_p = 800\ \mu s$		V_{F1}		2.75	V dc
Forward voltage		$t_p \leq 8.3$ ms, duty cycle $\leq 2\%$ pulse; $i_f = 50$ A(pk)		V_{F2}	---	1.40	v(pk)
Reverse current	4016	DC method V_R = rated V_R (see 1.3)		I_{R1}	---	15	µA dc
Subgroup 3							
High temperature operation:		$T_C = 150°C$					
Reverse current:	4016	DC method V_R = rated V_R (see 1.3)		I_{R2}	---	6.0	mA dc
Subgroup 4							
Reverse recovery time	4031	$I_F = 1$ A dc; $V_R = 30$ V dc; $di/dt = -25$ A/µs $i_r(rec) \leq 2$ A(pk) See 4.5.2 See figures 2 and 3					
Non A-versions				t_{rr}	---	200	ns
A-versions				t_{rr}	---	150	ns

1/ For LTPD numbers, see MIL-S-19500.

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TABLE II. Group B inspection for JANTX and JANTXV.

Inspection	MIL-STD-750		Qualification and large lot quality conformance inspection LTPD 1/	Small lot quality conformance inspection n/c 1/
	Method	Conditions		
Subgroup 1 2/				
Solderability	2026			
Resistance to solvents	1022			
Subgroup 2				
Thermal shock (temperature cycling)	1051			
Surge current	4066	T_C = 100°C; V_R = V_{RM}(see 1.3); six surges; 1 surge/minute max t_p = 8.3 ms		
Non-A version devices		i_f (surge) = 300 A dc I_O = 30 A dc		
A version devices		i_f (surge) = 400 A dc I_O = 50 A dc		
Hermetic seal	1071			
Fine leak Gross leak				
Electrical measurements		See table V, steps 1 and 2		
Subgroup 3 2/				
DC intermittent operation life	1037	.25 rated $I_O \leq I_O$ applied $\leq$ rated I_O (see 4.5.1) 2000 cycles		
Electrical measurements		See table V, steps 1, 2, and 3		
Steady-state dc blocking	1038 or 1049	Condition A, 340 hours		
Electrical measurements		See table V, steps 1 and 2		
Subgroups 4				
Decap internal visual (design verification)	2075			
Subgroup 5				
Not applicable				

See footnotes at end of table.

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TABLE II. Group B inspection for JANTX and JANTXV - Continued.

Inspection	MIL-STD-750		Qualification and large lot quality conformance inspection LTPD 1/	Small lot quality conformance inspection n/c 1/
	Method	Conditions		
Subgroup 6				
High-temperature (nonoperating)	1032	T_A = 175°C		
Electrical measurements		See table V, steps 1 and 2		

1/ For LTPD numbers, see MIL-S-19500.
2/ A separate sample may be pulled for each test.

TABLE III. Group C inspection.

Inspection	MIL-STD-750		Qualification and large lot quality conformance inspection LTPD 1/	Small lot quality conformance inspection n/c 1/
	Method	Conditions		
Subgroup 1				
Physical dimensions	2066	See figure 1		
Subgroup 2				
Thermal shock (glass strain)	1056			
Terminal strength	2036			
Bending stress		Test condition F method B; weight = 15 lbs. t = 15 s		
Seal torque		Test condition D1; Torque = 3 lb-in, t = 15 s		
Stud torque		Test condition D2; Torque - 30 lb-in; t = 15 s		
Hermetic seal	1071			
Fine leak Gross leak				

See footnote at end of table.

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TABLE III. Group C inspection - Continued.

Inspection	MIL-STD-750		Qualification and large lot quality conformance inspection LTPD 1/	Small lot quality conformance inspection n/c 1/
	Method	Conditions		
Subgroup 2 (cont'd)				
Moisture resistance	1021			
Electrical measurement		See table V, steps 1, 2, and 3		
Subgroup 3				
Shock	2016			
Vibration, variable frequency	2056			
Constant acceleration	2006			
Electrical measurements		See table V, steps 1 and 2		
Subgroup 4				
Salt atmosphere (corrosion)	1041			
Subgroup 5				
Not applicable				
Subgroup 6			5	12/0
DC intermittent operation life	1037	.25 rated $I_O \leq I_O$ applied $\leq$ rated I_O (see 4.5.1) 6000 cycles		
Electrical measurements		See table V, steps 1, 3, and 4		
Subgroup 7				
Not applicable				

1/ For LTPD numbers, see MIL-S-19500.

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TABLE IV. Group E inspection (all quality levels) for qualification only.

Inspection	MIL-STD-750		Qualification and large lot quality conformance inspection LTPD	Small lot quality conformance inspection n/c
	Method	Conditions		
Subgroup 1			10	8/0
Thermal shock (glass strain)	1056	0°C to 100°C, 100 cycles		
Hermetic seal	1071			
Electrical measurements		See table V, steps 1, 2, and 3		
Subgroup 2			5	12/0
Steady-state dc blocking life	1038	Condition A; 1000 hrs		
Electrical measurements		See table V, steps 1 and 4		
Subgroup 3		Photos of cross sections shall be submitted in the qualification report. Vendor shall retain duplicate photos.	3/0	1/0
DPA				
Subgroup 4			10	6/0
Thermal resistance	3101	$R_{\theta JC}$ = 0.8°C/W maximum I_M = 50 mA, I_H = 50 A, t_H = 400 ms, t_{md} = 80 µs		
Subgroup 5			10	6/0
Barometric pressure (reduced)	1001	Pressure (mm of Hg) / Voltage during test		
1N3912		8 / 300 V dc		
1N3913		8 / 400 V dc		
		I_R measurements on each device type shall not exceed group A limits.		

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TABLE V. Groups A, B, and C electrical end-point measurements.

Step	Inspection	MIL-STD-750		Symbol	Limits		Unit
		Method	Conditions		Min	Max	
1.	Forward voltage	4011	I_F = 50 A(pk) (pulse); pulse width = 8.3 ms; duty cycle = 2% maximum	V_{F2}	---	1.40	V(pk)
2.	Reverse current	4016	DC method V_R = Rated V_R (see 1.4)	I_{R1}	---	15	µA dc
3.	Thermal response	3101	See 4.3.2	ΔV_F			mV dc
4.	Reverse current	4016	DC method V_R = rated V_R (see 1.4)	I_{R3}	---	20	µA dc

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-S-19500.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

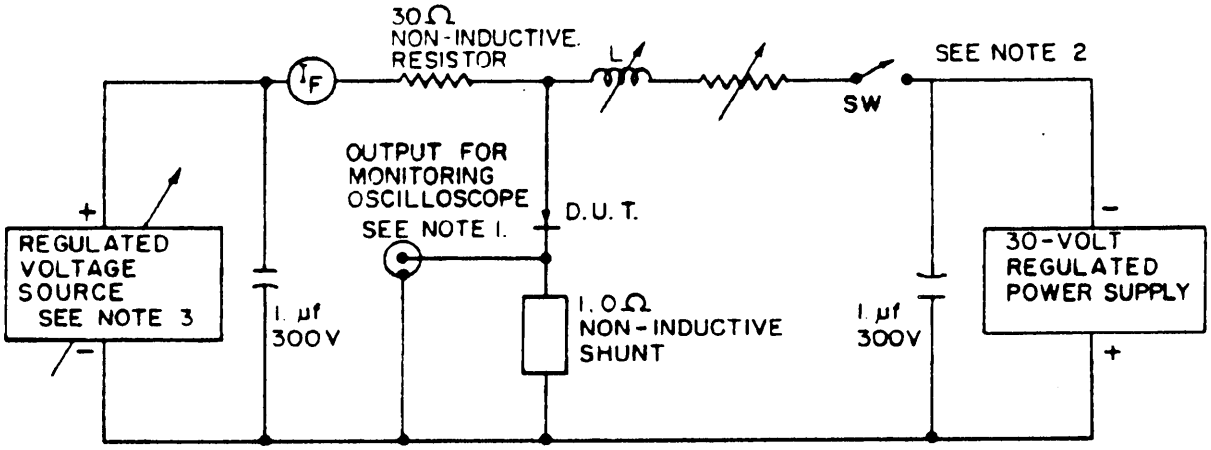
6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1).

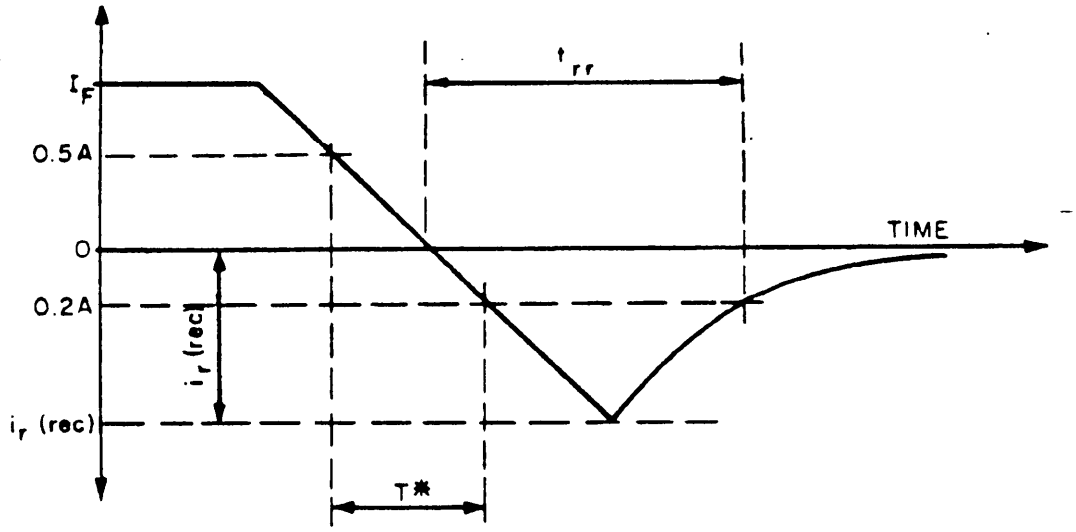
c. The lead finish may be specified.

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NOTES:
1. Monitoring oscilloscope requirements: $t_r \leq 14$ ns, $R_{in} \geq 9$ MΩ, $C_{in} \leq 12$ pF, L_{in} (series) ≤ 0.5 μH.
2. SW characteristics: Mercury-wetted make-before-break relay switched at a 60 Hz rate. The relay should conduct for approximately 640 μs and be open for approximately 7.7 ms. (C.P. Clare HGP 1004 or equivalent).
3. Voltage source characteristics: Output impedance $\leq$ 0.5Ω from 0 to 2 kHz.

FIGURE 2. Reverse recovery test circuit.



*Adjust L and R in accordance with paragraph 4.5.2 to achieve T = .028 μs (L = 1.2 μH). Then $\frac{di}{dt} = \frac{-0.7}{.028} = -25A/\mu s$.

FIGURE 3. Reverse recovery current waveform through device under test.

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6.3 Substitution information. Devices covered by this specification are substitutable for the manufacturers' and users' Part or Identifying Numbers (PIN). This information in no way implies that manufacturers' PIN's are suitable as a substitute for the military PIN.

Military PIN	Manufacturer's CAGE code	Manufacturer's and user's PIN

6.4 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 17

Review activities:
Army - AR, MI
Air Force - 11, 85
DLA - ES
NASA - NA

User activities:
Army - SM
Navy - AS, CG, MC, OS
Air Force - 19

Preparing activity:
Army - ER

Agent:
DLA - ES

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